THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

CITIZEN ALERT SYSTEM

Field	Value
Project Name	Citizen Alert System
Version	1.0
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Reviewed By	**MR MASSOUD HAMAD MMANGA**

1. Introduction

1.1 Purpose

This document describes how we will test the Citizen Alert System. It explains what we will test, how we will test it, and what tools we need. The goal is to make sure the system works correctly before people use it.

1.2 Scope

In Scope (What we WILL test):

- Office account creation and login
- Citizen registration and search
- Creating and viewing notifications
- Notification targeting (by region, status, education)
- Priority system (Low, Normal, High, Emergency)
- Super Admin functions (manage accounts)

Out of Scope (What we will NOT test):

- Mobile app version (not available yet)
- SMS/email delivery (not implemented)
- Network security (basic only)
- Load testing with 1000+ users

1.3 Testing Objectives

1. Make sure all menus and options work correctly
2. Check that data is saved and retrieved properly
3. Confirm only authorized users can access features
4. Find and fix bugs before release
5. Ensure citizens see correct alerts based on their profile

2. Test Strategy

2.1 Testing Levels

2.1.1 Unit Testing

- **Objective:** Test each function works alone
- **Responsibility:** AbdulKadir Bakar Mbwana (Developer)
- **What we'll test:** Login function, search function, add notification function
- **When:** During coding

2.1.2 Integration Testing

- **Objective:** Test if functions work together
- **Responsibility:** Khadija Ali Said (System Analyst)
- **What we'll test:** Login → Create notification → View notification
- **When:** After all functions are coded

2.1.3 System Testing

- **Objective:** Test the complete system
- **Responsibility:** Entire team
- **What we'll test:** All features together, like real users would use them
- **When:** Before handing to users

2.1.4 User Acceptance Testing (UAT)

- **Objective:** Let real users try the system
- **Responsibility:** Zahor Khalfan Omar (Project Manager) with selected users
- **What we'll test:** Can they register? Can offices send alerts?
- **When:** Final stage before completion

2.2 Testing Types

Test Type	What It Means	Priority
Functional Testing	Check if buttons and menus do what they should do.	High
Usability Testing	Is it easy to understand and use?	Medium
Security Testing	Can someone access without login?	High
Regression Testing	After fixing bugs, check if other things broke	High
Edge Case Testing	Try empty inputs, wrong inputs, long text	Medium

3. Test Environment

3.1 Hardware Requirements

- **Computer:** Any laptop/desktop (Windows, Mac, or Linux)
- **RAM:** 4GB minimum
- **Storage:** 100MB free space

3.2 Software Requirements

- **Operating System:** Windows 10/11, Ubuntu 20.04+, or macOS
- **Compiler:** GCC (to compile the C program)
- **Terminal:** Command prompt or terminal application

3.3 Test Tools

Tool	What We Use It For
GCC Compiler	To build the program from source code
Git/GitHub	Track changes and manage code versions
Pen and Paper	Note bugs and issues during testing
Excel/Google Sheets	Record test results
WhatsApp/Discord	Team communication

4. TEST CASES

4.1 Test Case Template

We will use this simple format for all tests:

Field	Description
Test ID	Unique number for the test
Test Title	Short name describing the test
Module	Which part of the system (Login, Notifications, etc.)
Steps	What to do step-by-step
Expected Result	What should happen
Actual Result	What actually happened (fill during testing)
Status	Pass / Fail / Blocked

4.2 Authentication Test Cases (Office Login).

TC-AUTH-001

Test Title	First time login - create Super Admin account
Module	Office Authentication
Steps	1. Run the program 2. Choose Office Admin (0) 3. When prompted, create username, password, office name
Expected Result	System creates first account as Super Admin and asks to login
Actual Result	
Status	Pending

TC-AUTH-002

Test Title	Login with correct credentials
Module	Office Authentication
Steps	1. Choose Office Admin (0) 2. Enter correct username and password
Expected Result	Login successful - shows welcome message with office name and role
Actual Result	
Status	Pending

TC-AUTH-003

Test Title	Login with wrong password
Module	Office Authentication
Steps	1. Choose Office Admin (0) 2. Enter correct username but wrong password
Expected Result	"Login failed: account not found" message appears
Actual Result	
Status	Pending

TC-AUTH-004

Test Title	Login with non-existent username
Module	Office Authentication
Steps	1. Choose Office Admin (0) 2. Enter username that doesn't exist
Expected Result	"Login failed: account not found" message appears
Actual Result	
Status	Pending

4.3 Super Admin Test Cases (Account Management)

TC-SUPER-001

Test Title	Create new Admin account
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 5 (Create Office Account) 3. Enter username, password, office name 4. Choose role 0 (Admin)
Expected Result	Account created successfully with ID shown
Actual Result	
Status	Pending

TC-SUPER-002

Test Title	List all office accounts
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 6 (List Office Accounts)
Expected Result	Table showing all accounts with ID, username, office, role, active status
Actual Result	
Status	Pending

TC-SUPER-003

Test Title	Deactivate an account
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 7 (Toggle Account Active) 3. Enter an account ID
Expected Result	Account status changes from Active to Inactive (or vice versa)
Actual Result	
Status	Pending

TC-SUPER-004

Test Title	Delete another admin account
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 8 (Delete Office Account) 3. Enter another admin's ID
Expected Result	Account deleted successfully
Actual Result	
Status	Pending

TC-SUPER-005

Test Title	Try to delete own account
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 8 (Delete Office Account) 3. Enter your own ID
Expected Result	"Cannot delete your own account" message appears
Actual Result	
Status	Pending

TC-SUPER-006

Test Title	Reset another admin's password
Module	Super Admin - Account Management
Steps	1. Login as Super Admin 2. Choose option 9 (Reset Account Password) 3. Enter another admin's ID and new password
Expected Result	Password updated successfully
Actual Result	
Status	Pending

4.4 Notification Test Cases (For All Offices)

TC-NOTIF-001

Test Title	Create notification - Target All
Module	Notifications
Steps	1. Login as any office 2. Choose option 1 (Create Notification) 3. Enter title, message 4. Choose priority (1=Normal) 5. Choose target 0 (All)
Expected Result	Notification created with ID shown
Actual Result	
Status	Pending

TCNOTIF002

Test Title	Create notification - Target Region
Module	Notifications
Steps	1. Login as any office 2. Choose option 1 (Create Notification) 3. Enter title, message 4. Choose priority (2=High) 5. Choose target 1 (Region) 6. Enter region name (e.g., "Unguja")
Expected Result	Notification created with region target
Actual Result	
Status	Pending

TC-NOTIF-003

Test Title	Create notification - Target Status
Module	Notifications
Steps	1. Login as any office 2. Choose option 1 (Create Notification) 3. Enter title, message 4. Choose priority (3=Emergency) 5. Choose target 2 (Status) 6. Enter status (0=Student)
Expected Result	Notification created with status target
Actual Result	
Status	Pending

TC-NOTIF-004

Test Title	Create notification - Target Education
Module	Notifications
Steps	1. Login as any office 2. Choose option 1 (Create Notification) 3. Enter title, message 4. Choose priority (1=Normal) 5. Choose target 3 (Education) 6. Enter education level (2=A-Level)
Expected Result	Notification created with education target
Actual Result	
Status	Pending

TC-NOTIF-005

Test Title	List notifications (Admin)
Module	Notifications
Steps	1. Login as regular Admin 2. Choose option 2 (List Notifications)
Expected Result	Shows only notifications from this admin's office, sorted by priority
Actual Result	
Status	Pending

TC-NOTIF-006

Test Title	List notifications (Super Admin)
Module	List notifications (Super Admin)
Steps	1. Login as Super Admin 2. Choose option 2 (List Notifications)
Expected Result	Shows ALL notifications from ALL offices
Actual Result	
Status	Pending

TC-NOTIF-007

Test Title	Search notification by ID
Module	Notifications
Steps	1. Login as any office 2. Choose option 3 (Search Notification) 3. Enter a valid notification ID
Expected Result	Shows full details of that notification
Actual Result	
Status	Pending

TC-NOTIF-008

Test Title	Remove own office notification (Admin)
Module	Notifications
Steps	1. Login as regular Admin 2. Choose option 4 (Remove Notification) 3. Enter ID of notification from your office
Expected Result	Notification marked as inactive (removed)
Actual Result	
Status	Pending

TC-NOTIF-009

Test Title	Try to remove other office notification (Admin)
Module	Notifications
Steps	1. Login as regular Admin 2. Choose option 4 (Remove Notification) 3. Enter ID of notification from different office
Expected Result	"Access denied" message appears
Actual Result	
Status	Pending

TC-NOTIF-010

Test Title	Remove any notification (Super Admin)
Module	Notifications
Steps	1. Login as Super Admin 2. Choose option 4 (Remove Notification) 3. Enter any notification ID
Expected Result	Notification marked as inactive
Actual Result	
Status	Pending

4.5 Citizen Test Cases

TC-CIT-001

Test Title	Register new citizen
Module	Citizen Portal
Steps	1. Choose Citizen (1) from main menu 2. Choose option 1 (Register) 3. Enter name, region 4. Choose status (1=Worker) 5. Choose education (4=Undergrad)
Expected Result	Citizen registered with ID shown
Actual Result	
Status	Pending

TC-CIT-002

Test Title	Search citizen by ID
Module	Citizen Portal
Steps	1. Choose Citizen (1) 2. Choose option 2 (Search Citizen) 3. Choose 1 for ID search 4. Enter valid citizen ID
Expected Result	Shows citizen details (name, region, status, education)
Actual Result	
Status	Pending

TC-CIT-003

Test Title	Search citizen by name
Module	Citizen Portal
Steps	1. Choose Citizen (1) 2. Choose option 2 (Search Citizen) 3. Choose 2 for name search 4. Enter part of a citizen's name
Expected Result	Shows matching citizen(s)
Actual Result	
Status	Pending

TC-CIT-004

Test Title	View my alerts - Citizen sees matching notifications
Module	Citizen Portal
Steps	1. Choose Citizen (1) 2. Choose option 3 (View My Alerts) 3. Enter your citizen ID
Expected Result	Shows only notifications that match your profile (region, status, education)
Actual Result	
Status	Pending

TC-CIT-005

Test Title	View my alerts - Emergency notifications appear for everyone
Module	Citizen Portal
Preconditions	There is an Emergency priority notification
Steps	1. Choose Citizen (1) 2. Choose option 3 (View My Alerts) 3. Enter any citizen ID
Expected Result	Emergency notification shows even if it doesn't match profile
Status	Pending

4.6 Edge Cases Test Cases

TC-EDGE-001

Test Title	Create notification with empty title
Module	Notifications
Steps	1. Login as office 2. Create notification 3. Press Enter without typing title
Expected Result	Should accept empty title? (Need to check behavior)
Actual Result	
Status	Pending

TC-EDGE-002

Test Title	Enter letters when number is expected
Module	All modules
Steps	1. At any menu asking for number (priority, target, etc.) 2. Type letters instead of numbers
Expected Result	Program should handle gracefully, not crash
Actual Result	
Status	Pending

TC-EDGE-003

Test Title	Very long text input
Module	All modules
Steps	1. When asked for name or message 2. Type 100+ characters
Expected Result	Program should handle without crashing
Actual Result	
Status	Pending

TC-EDGE-004

Test Title	Search for non-existent ID
Module	All modules
Steps	1. Search for citizen or notification 2. Enter ID that doesn't exist
Expected Result	"Not found" message appears
Actual Result	
Status	Pending

Project Repository: https://github.com/zvhor/citizen_alert_system.git

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